Basic formulas:

$$a^0 = 1$$

$$a^1 = a$$

$$a^{-n} = \frac{1}{a^n}$$

$$a^{\frac{1}{n}} = \sqrt[n]{a}$$

$3^\circ = 1$

$0^5 = 0$

$0^\circ \rightarrow$ Not defined

$a^{\frac{1}{2}} = \sqrt{a}$

$a^{\frac{1}{3}} = \sqrt[3]{a}$

$\rightarrow \dfrac{5^{-x}}{7^{2x-y}} = \dfrac{7^{y-2x}}{5^x}$

coaching center

Basic formulas:

$$a^m \times a^n = a^{m+n}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$(a^m)^n = a^{m \times n}$$

$$a^{m^n} = a^{(m^n)^p}$$

$$(ab)^n = a^n \times b^n$$

Basic formulas:

$1^5 = 1^x$

$2^{20} = x^{20}$ → $2 \mid -2$

$2^5 = x^5$ → ②

If $a^m = a^n$, then

$m = n$ $(a \neq 0, 1, -1)$

$5^7 = 5^x \Rightarrow 7 = x$

$0^{1000} = 0^{500} \Rightarrow 1000 = 500$ ✗

$1^{1000} = 1^{500}$

If $a^m = b^m$, then

$a = \pm b$, if m is even

$a = b$, if m is odd

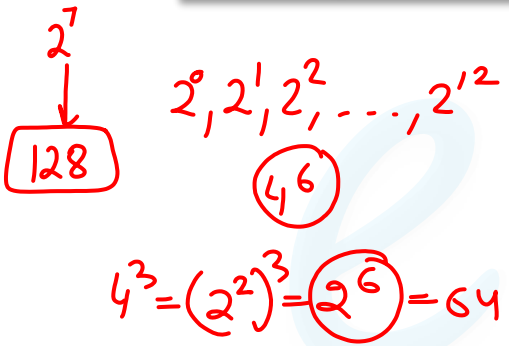
$2^{10} = x^{10}$ $\quad 2 = x$

$= 2^{10}$

$= (-2)^{10}$

coaching centers

Things to memorize:



- $2^{12}, 3^6, 5^5, 6^4$
- Squares up to 30
- Cubes up to $12, (15, 21)$
- $\sqrt{2}, \sqrt{3}, \sqrt{5}$

1.414 1.732 2.236

coaching center

① Recognize the rational number:

परिमेय संख्या बताइए:

a) $\sqrt[4]{27}$ b) $\sqrt[3]{16}$ c) $\sqrt{11}$ d) $\sqrt[5]{243}$

करणी / Surds → $\sqrt{2}$, $\sqrt[3]{4}$, $\sqrt[5]{16}$ ← Irrational

Rational no → $\sqrt{25}$ = 5

coaching center

② Which of the following is a surd?

निम्न में से कौनसी करणी/अपरिमेय संख्या है?

a) $\sqrt[4]{81}$ b) $\sqrt[5]{32}$ c) $\sqrt[6]{729}$ d) $\sqrt[3]{25}$

e1 coaching center

③ $\left[\sqrt[14]{\sqrt[3]{\left(\sqrt[5]{\left(\sqrt[3]{5^6}\right)^2}\right)^{15}}}\right]^7$

a) 5^2 b) 5^4 c) 5^8 d) 5^{12}

$\left(\left(\left(a^m\right)^n\right)^p\right)^q = a^{m\,n\,p\,q} = a^{m\times p\times n\times q}$

$= 5^{6\times 2\times 15\times 7\times \frac{1}{3}\times \frac{1}{5}\times \frac{1}{5}\times \frac{1}{14}}$

$= 5^2$

coaching center

4) $\frac{\left((xy)^{a-b}(xy)^{b-c}\right)}{(xy)^{a-c}} = ?$

a) 1 b) 0 c) xy d) xy^{abc}

$$\frac{xy^{a-c}}{xy^{a-c}} = 1$$

e1 coaching center

$\frac{a^m}{a^n} = a^{m-n}$

⑤ $\left(\frac{x^a}{x^b}\right)^{a+b} \times \left(\frac{x^b}{x^c}\right)^{b+c} \times \left(\frac{x^c}{x^a}\right)^{c+a} = ?$

a) 1 b) 0 c) x d) x^{abc}

$(x^{a-b})^{a+b}$

$= x^{a^2-b^2} \times x^{b^2-c^2} \times x^{c^2-a^2}$

$= x^0 = 1$

e1 coaching center

⑥ The value of $\left(\frac{x^a}{x^b}\right)^{\frac{1}{ab}} \cdot \left(\frac{x^b}{x^c}\right)^{\frac{1}{bc}} \cdot \left(\frac{x^c}{x^a}\right)^{\frac{1}{ca}} = ?$

$\left(x^{a-b}\right)^{\frac{1}{ab}}$

$\left(\frac{x^a}{x^b}\right)^{\frac{1}{ab}} \cdot \left(\frac{x^b}{x^c}\right)^{\frac{1}{bc}} \cdot \left(\frac{x^c}{x^a}\right)^{\frac{1}{ca}}$ का मान है :

a) 1 b) $\frac{1}{x^{abc}}$ c) $\frac{1}{x^{ab+bc+ca}}$ d) None of these

$\frac{a-b}{ab} = \frac{a}{ab} - \frac{b}{ab}$

$= x^{\frac{a-b}{ab}} \; x^{\frac{b-c}{bc}} \; x^{\frac{c-a}{ca}}$

$= x^{\frac{1}{b}-\frac{1}{a}} \cdot x^{\frac{1}{c}-\frac{1}{b}} \; x^{\frac{1}{a}-\frac{1}{c}}$

$= x^0$

coaching center

7) $4^{4x+1} = \frac{1}{64}$, then x is

HW a) $\frac{1}{2}$ b) -1 c) $-\frac{1}{2}$ d) $-\frac{1}{6}$

$4^{4x+1} = 4^{-3}$

$\Rightarrow 4x+1 = -3$

$\Rightarrow x = \frac{-4}{4} = -1$

coaching center

(8) If $5^{5x+5} = 1$, then x equals

HW a) 0 b) -1 c) 1 d) $-\frac{4}{5}$

$5^{5x+5} = 5^0$

$\Rightarrow 5x+5 = 0$

$\Rightarrow x = -1$

coaching center

⑨ If $27^{2x-1} = 243^3$, then x equals

~~a) 3~~ b) 6 c) 7 d) 9

$(a^m)^n = a^{mn}$

$3^{3(2x-1)} = 3^{5\times3}$

$2x = 6$

$x = 3$

coaching center

(10) If $8^{3x-5} = \frac{1}{32^{7-4x}}$ then $x = ?$

अगर $8^{3x-5} = \frac{1}{32^{7-4x}}$ है तो $x = ?$

a) $\frac{16}{9}$ b) $\frac{20}{11}$ c) $\frac{25}{13}$ d) 2

$\frac{1}{a^m} = a^{-m}$

$2^{9x-15} = 2^{20x-35}$

$20 = 11x$

coaching center

(11) If $\left(\frac{x}{y}\right)^{5a-3} = \left(\frac{y}{x}\right)^{17-3a}$ then a =?

अगर $\left(\frac{x}{y}\right)^{5a-3} = \left(\frac{y}{x}\right)^{17-3a}$ है तो a =?

a) -5 b) -6 c) -7 d) 7

$\frac{1}{a^m} = a^{-m}$

$\left(\frac{1}{a}\right)^m = \left(\frac{a}{1}\right)^{-m}$

$\left(\frac{x}{y}\right)^a = \left(\frac{y}{x}\right)^{-a}$

$5a-3 = 3a-17$

$2a = -14$

$a = -7$

e1 coaching center

(12) $x^{x\sqrt{x}} = (x\sqrt{x})^{x}$, then x equals

$x^{x\sqrt{x}} = (x\sqrt{x})^{x}$ है तो x का मान है

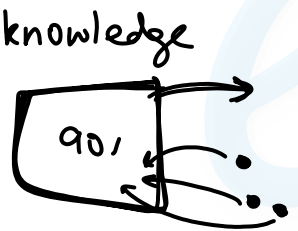
a) $\frac{4}{9}$ b) $\frac{2}{3}$ c) $\frac{9}{4}$ d) $\frac{3}{2}$

$$\left(x^{\frac{3}{2}}\right)^{x}$$

$$x^{x\sqrt{x}} = x^{\frac{3}{2}x}$$

$$\Rightarrow x\sqrt{x} = \frac{3}{2}x$$

$$x = \frac{9}{4}$$



Mock

Ques X

Concept

coaching center

(13) If $(x^x)^{\frac{5}{4}} = x^{x^{\frac{5}{4}}}$ then x =?

अगर $(x^x)^{\frac{5}{4}} = x^{x^{\frac{5}{4}}}$ है तो x =?

a) $\frac{125}{64}$ b) $\frac{625}{256}$ c) $\frac{25}{16}$ d) $\frac{5}{4}$

$x^{\frac{5}{4}x} = x^{\left(x^{\frac{5}{4}}\right)}$

$\Rightarrow \frac{5}{4}x^1 = x^{5/4}$

$\frac{5}{4} = x^{\frac{1}{4}}$

$\frac{625}{256} = x$

1. knowledge
2. Practice / Concepts

coaching center

(14) Simplify $(5 \times 5 \times 5 \times 5 \times 5)^5 \times (5 \times 5 \times 5)^5 \div 5 = (125)^?$

$(5 \times 5 \times 5 \times 5 \times 5)^5 \times (5 \times 5 \times 5)^5 \div 5 = (125)^x$ को सरल करें|

a) 15 b) 13 c) 21 d) 14

$$\frac{(5^5)^5 \times (5^3)^5}{5^1} = 5^{3x}$$

$$5^{13} = 5^{3x}$$

coaching center

15 Find the value of $\left(\frac{9^3+9^4+9^5+9^6+9^7}{9^1+9^2+9^3+9^4+9^5}\right)^{\frac{1}{2}}$

$\left(\frac{9^3+9^4+9^5+9^6+9^7}{9^1+9^2+9^3+9^4+9^5}\right)^{\frac{1}{2}}$ का मान ज्ञात करें|

a) 81 b) 9 c) 729 d) 3

$\left(9^2\left(\frac{\quad}{\quad}\right)\right)^{\frac{1}{2}}$

$= 9$

coaching center

16 If $(3^{33} + 3^{33} + 3^{33})(2^{33} + 2^{33}) = 6^x$, then what is the value of x?

यदि $(3^{33} + 3^{33} + 3^{33})(2^{33} + 2^{33}) = 6^x$ हो, तो x का क्या मान है?

a) 34 b) 35 c) 33 d) 33.5

$x+x+x=3x$

$(3^1 \cdot 3^{33})$ $(2 \cdot 2^{33})$

$3^{34}\, 2^{34} = 6^x$

$\Rightarrow 6^{34} = 6^x$

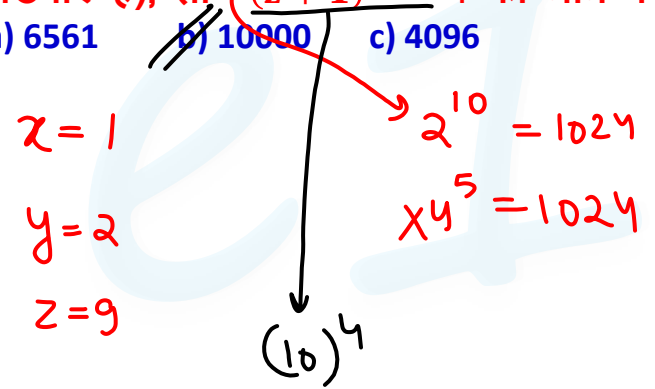
$a^n b^n = (ab)^n$

coaching center

(17) If $x^{y+Z} = 1, y^{x+z} = 1024$ *and* $z^{x+y} = 729$ (x, y and z are natural number), then what is the value of $(z+1)^{y+x+1}$?

यदि $x^{y+Z} = 1, y^{x+z} = 1024$ *and* $z^{x+y} = 729$(x,y तथा z प्राकृतिक संख्याएं है), तो $(z+1)^{y+x+1}$? का मान क्या है?

a) 6561 b) 10000 c) 4096 d) 14641



coaching center

(18) If $a = 0.4039$ then $\sqrt{4a^2 - 4a + 1} + 5a = ?$

अगर $a = 0.4039$ है तो $\sqrt{4a^2 - 4a + 1} + 5a$ का मान:

a) 2.2117 b) 2.4039

c) 2.8078 d) 1.8273

$1-2a+5a$

$= 1+3a$

$= 1+1.2117$

$\sqrt{(2a-1)^2}$ $\sqrt{(1-2a)^2}$

$\times$ $2a-1$ $1-2a$ ✓

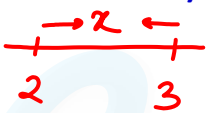
$-ve$

coaching center

(19) If $2 < x < 3$ then $\sqrt{(2-x)^2} + \sqrt{(x-3)^2} = ?$

अगर $2 < x < 3$ है तो $\sqrt{(2-x)^2} + \sqrt{(x-3)^2} = ?$

a) 1 b) -1 c) $2x - 5$ d) $5 - 2x$



$x - 2 + 3 - x = 1$

$\sqrt{\quad}$ → +ve

coaching center

20 If $x = y^a, y = z^b$ and $z = x^c$, then the value of abc is

यदि $x = y^a, y = z^b$ और $z = x^c$ है,तो abc का मान है:

a) 1 b) 2 c) -1 d) 0

$y = (x^c)^b = x^{bc}$

$x^1 = (x^{bc})^a = x^{abc}$

coaching center

(21) If $a^x = b, b^y = c$ and $xyz = 1$, then what is the value of c^z?

यदि $a^x = b, b^y = c$ और $xyz = 1$ है, तो c^z का मान क्या होगा ?

a) a b) b c) ab d) $\frac{a}{b}$

$$(a^x)^y = c$$

$$(a^{xy})^z = c^z$$

$$a = a^{xyz} = c^z$$

coaching center

(22) If $a^x = b^y = c^z$ and $abc = 1$, then the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ will be equal to

यदि $a^x = b^y = c^z$ और $abc = 1$ है, तो $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ का मान बराबर है :

a) -1 b) 0 c) 1 d) 3

$a^x = b^y = c^z = k$

$c^z = k$ $\quad b^y = k$ $\quad a^x = k$

$c = k^{\frac{1}{z}}$ $\quad b = k^{\frac{1}{y}}$ $\quad a = k^{\frac{1}{x}}$

$k^{\frac{1}{x}} \, k^{\frac{1}{y}} \, k^{\frac{1}{z}} = 1$

$k^{\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)} = k^0$

coaching center

(23) If $a^{\frac{1}{m}} = b^{\frac{1}{n}} = c^{\frac{1}{p}}$ and $abc = 1$, then $(m + n + p)$ is equal to

यदि $a^{\frac{1}{m}} = b^{\frac{1}{n}} = c^{\frac{1}{p}}$ और $abc = 1$ है, तो $(m + n + p)$ बराबर है:

a) 0 b) 2 c) 1 d) -2

$a^{\frac{1}{m}} = k$

$a = k^m$

$b = k^n$

$c = k^p$

$k^m \, k^n \, k^p = 1$

$k^{m+n+p} = 1 = k^0$

coaching center

(24) If $2^x = 3^y = 6^{-z}$ then $\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$ is

अगर $2^x = 3^y = 6^{-z}$ है तो $\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$=?

$\frac{1}{x}+\frac{1}{y}=-\frac{1}{z}$

$=k$

a) 0 b) 1 c) $\frac{3}{2}$ d) $-\frac{1}{2}$

$2 = k^{\frac{1}{x}}$

$3 = k^{\frac{1}{y}}$

$6 = k^{\frac{-1}{z}}$

$2\times3=6$

$k^{\frac{1}{x}} \times k^{\frac{1}{y}} = k^{\frac{-1}{z}}$

$k^{\frac{1}{x}+\frac{1}{y}} = k^{\frac{-1}{z}}$

$\frac{1}{x}+\frac{1}{y}=-\frac{1}{z}$

coaching center

(25) If $4^x = 6^{-y} = 9^z$ then $\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$ is

अगर $4^x = 6^{-y} = 9^z$ है तो $\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$=?

a) 0 b) $-\frac{1}{x}$ c) $-\frac{1}{y}$ d) $-\frac{1}{z}$

$4 = k^{\frac{1}{x}}$

$6^2 = k^{-\frac{2}{y}}$

$9 = k^{\frac{1}{z}}$

$6^2 = 4 \times 9$

$k^{-\frac{2}{y}} = k^{\frac{1}{x}} k^{\frac{1}{z}}$

$-\frac{2}{y} = \frac{1}{x} + \frac{1}{z} + \frac{1}{y}$

coaching center

(26) If $2^x = 3^y = 12^z$ then $\left(\frac{1}{z} - \frac{1}{y}\right)$ is

अगर $2^x = 3^y = 12^z$ है तो $\left(\frac{1}{z} - \frac{1}{y}\right)$=?

a) 0 b) $\frac{1}{x}$ c) $\frac{2}{x}$ d) $\frac{3}{x}$

$= k$

$2 = k^{\frac{1}{x}}$

$3 = k^{\frac{1}{y}}$

$12 = k^{\frac{1}{z}}$

$2^2 \times 3 = 12$

$k^{\frac{2}{x}} \times k^{\frac{1}{y}} = k^{\frac{1}{z}}$

$\frac{2}{x} + \frac{1}{y} = \frac{1}{z}$ $-\frac{1}{y}$

coaching center

(27) If $(3.7)^x = (0.037)^y = 10000$, then what is the value of $\frac{1}{x} - \frac{1}{y}$?

यदि $(3.7)^x = (0.037)^y = 10000$ है, तो $\frac{1}{x} - \frac{1}{y}$ का मान क्या होगा ?

a) 1 b) 2 c) $\frac{1}{2}$ d) $\frac{1}{4}$

10^4

$\frac{100}{10}$

$3.7 = (10^4)^{\frac{1}{x}}$

$.037 = (10^4)^{\frac{1}{y}}$

$\frac{a^m}{a^n} = a^{m-n}$

$100 = (10^4)^{\frac{1}{x} - \frac{1}{y}}$

$\Rightarrow 10^2 = 10^{4\left(\frac{1}{x} - \frac{1}{y}\right)}$

$2 = 4\left(\frac{1}{x} - \frac{1}{y}\right)$

coaching center

$$\frac{(.4)^{369}}{(.4)^{94}} \times .4^{63} = (.4)^{x}$$

$$.4^{338} = .4^{x}$$

28 **Select the number that will come in place of the question mark (?) in the mathematical statement.**

$(0.064)^{123} \div 0.16^{47} \times 0.4^{34} \times 0.4^{29} = (0.4)^{?}$

निम्न समीकरण में प्रश्न चिह्न (?) के स्थान पर कौन-सी संख्या आ सकती है?

$(0.064)^{123} \div 0.16^{47} \times 0.4^{34} \times 0.4^{29} = (0.4)^{?}$

a) 350 **b) 320** **c) 338** **d) 341**

coaching center

(29) If $A = \frac{\sqrt{0.0004} \times \sqrt[3]{0.000008}}{\sqrt[4]{16000} \times \sqrt[3]{125000} \times \sqrt[4]{810}}$ and $B = \frac{\sqrt[3]{0.729} \times \sqrt[4]{0.0016}}{\sqrt{0.16}}$, then what is $A \times B$?

अगर $A = \frac{\sqrt{0.0004} \times \sqrt[3]{0.000008}}{\sqrt[4]{16000} \times \sqrt[3]{125000} \times \sqrt[4]{810}}$ और $B = \frac{\sqrt[3]{0.729} \times \sqrt[4]{0.0016}}{\sqrt{0.16}}$ तो $A \times B$ क्या है?

a) 7×10^{-7}

b) $\left(\frac{7}{4}\right) \times 10^{-8}$

c) 5×10^{-8}

d) $\left(\frac{7}{3}\right) \times 10^{-7} \times \frac{3}{5}$

$\sqrt[y]{a} \times \sqrt[y]{b} = \sqrt[y]{ab}$

$$\frac{.02 \times .02 \times .9 \times .2}{2 \times 3 \times 10 \times 5 \times 10 \times .4} = \frac{3}{5} \times 10^{-7}$$

coaching center

Solve $\sqrt{2\sqrt{2}}$ $= 2^{\frac{1}{2}\times\frac{1}{2}} \times 2^{\frac{1}{2}} = 2^{\frac{1}{4}+\frac{1}{2}}$

Solve $\sqrt{2\sqrt{2\sqrt{2}}}$ $= 2^{\frac{1}{8}+\frac{1}{4}+\frac{1}{2}}$

Solve $\sqrt{6\sqrt{2\sqrt{3}}}$ 2×3 $3^{\frac{1}{8}+\frac{1}{2}} \times 2^{\frac{1}{4}+\frac{1}{2}}$

Solve $\sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}$ $2^{\frac{1}{16}+\frac{1}{8}+\frac{1}{4}+\frac{1}{2}}$

Solve $\sqrt{4\sqrt{8\sqrt{16}}}$ $2^{4\times\frac{1}{8}}$ $2^{3\times\frac{1}{4}}$ $2^{2\times\frac{1}{2}}$

coaching centers